

Sojeong Shin

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1 Education

- **Master of Science in Computer Science**
The State University of New York, Stony Brook | Expected Graduation Date: June 2026 | GPA: 3.89/4.0
- **Bachelor of Science in Computer Science and Engineering**
Incheon National University, Republic of Korea | Graduation Date: August 2024 | GPA: 3.62/4.5

2 International Experience

- Exchange Student at California State University / Fall semester, 2022
- Volunteering and ESL Education at GRA Korea in Singapore and Malaysia / January 2016

3 Professional Experience

1) THE STATE UNIVERSITY OF NEWYORK, STONYBROOK (Graduate School)

All You Need is Dictionary: Dictionary-assisted LLM fine-tuning (Expected to be submitted in May 2026)

- To upgrade the contextual understanding of LLMs without relying on massive additional datasets, this study proposes a lightweight fine-tuning technique that directly integrates dictionary definitions into the model.
- By updating only the embedding layer via an attention mechanism, the approach seamlessly aligns the model's internal semantic space with the actual dictionary sense space while preserving its pre-trained knowledge.
- This data-efficient method achieved a 6% performance improvement on the Semcor benchmark using only about 50MB of additional data, demonstrating its effectiveness in refining language models even in resource-constrained environments.

Monocular Camera-based 3D Measurement System (Provost's Award, Deployed Product)

- Developed a mathematically grounded geometric algorithm to estimate vertical distances using a single RGB camera, addressing real-world constraints where depth sensors or multi-view setups are infeasible.
- Integrated open-source computer vision frameworks (OpenCV, MediaPipe, DeepLab, etc.) and translated heuristic assumptions into quantitative formulations to enable validation and domain adaptability.
- Successfully productized and deployed the system through a startup collaboration with MOIDA; the solution is actively used in South Korean subway stations for public human exercise services.
- Won Provost's Award in SUNY research challenge in 2026.

Satellite-based Urban Heat Island Analysis Dashboard (Startup Collaboration)

- Built an end-to-end satellite imagery analysis system to study urban heat island effects in Central Asia, focusing on climate-driven environmental change and energy-related impacts.
- Designed a scalable geospatial pipeline using Google Earth Engine (GEE) and Google Cloud Storage (GCS) for large-scale satellite data processing and temporal analysis.
- Delivered results as an interactive web-based dashboard, enabling energy and environmental stakeholders to support data-driven decisions for mitigation strategies and sustainable infrastructure planning.

2) INCHEON NATIONAL UNIVERSITY, INCHEON (Undergraduate School)

Outstanding Graduation Project: AI personal training service in VR space (2nd Place Award)

Presentation Video: <https://youtu.be/OlViYYn-kFo>

- Designed and implemented a single-camera-based VR workout system using KalidoKit and mediapipe, enabling immersive exercise experiences without specialized sensors.
- Achieved accurate posture estimation using Euler rotation, quaternion mathematics, and linear regression, emphasizing mathematical rigor over hardware dependency.
- Led structured team collaboration using GitHub, Figma, Discord, and Notion, and received second place in the annual graduation project competition.

AWS Glocal Hackathon Participant

- Participated in the AWS Glocal Hackathon, collaborating with students from five different universities in a time-constrained, high-pressure environment.
- Led the development of “God Road,” a parking-sharing platform designed to address local urban parking challenges as a team leader.
- Demonstrated leadership and cross-team collaboration using GitHub and Slack for agile development and project coordination.

AI Competition: Real-time Posture Correction on mobile device (1st Place Award)

- Developed a real-time posture correction system using a smartphone camera to address forward head posture and lumbar disc health issues.
- Extracted 3D human pose coordinates via MediaPipe and trained a One-Class SVM (OCSVM) model for abnormal posture detection.
- Awarded first place for proposing an innovative AI-driven healthcare solution and demonstrating effective teamwork.

Data Mining: Analysis and Forecasting of Solar Energy Production in Calgary

- Processed and analyzed large-scale real-world solar energy datasets, focusing on temporal patterns and noise characteristics in energy production data.
- Applied LOESS smoothing and FFT-based decomposition to separate trend, seasonality, and frequency components for interpretability.
- Conducted time-series forecasting using ACF, PACF, and Holt–Winters methods, achieving accurate short-term energy production predictions.

Machine Learning: Audio Classification & Noise Reduction Model

- Diagnosed a performance bottleneck during a machine learning voice classification project, identifying that ambient noise limited the model's accuracy to 50%.
- Engineered a customized preprocessing pipeline by analyzing the physical frequency characteristics of the audio data, utilizing Gaussian filtering and spectrogram transformations to extract optimal features for the CNN model.
- Achieved a 98.8% final classification accuracy and secured 1st place in the course project by successfully training the model on noise-robust, meaningful patterns.

3) INDIVIDUAL INSTRUCTOR

Robotics and Machine Learning Boot Camp TA / Incheon Academy of Science and Arts (IASA) / July 2025

- Led practical lab sessions as a teaching assistant for introductory Machine Learning courses covering transfer learning and CNN-based models, as well as Robotics courses with a focus on reinforcement learning.

English Conversation Instructor / ICME / September 2023 ~ April 2024

- Processed and analyzed large-scale real-world solar energy datasets, focusing on temporal patterns and noise characteristics in energy production data.

Programming Language Instructor / C-Cube Coding Institution / January 2022 ~ August 2022

- Taught Python, C, and Scratch to diverse groups of students implementing structured lessons.

4 Skills

- Programming Languages : Python, MATLAB, C, HTML
- Machine Learning Frameworks : Pytorch, HuggingFace, Transformer, TensorFlow, Keras, Scikit-Learn
- Vision Tools : Event Camera, RGB Camera, Mediapipe, MetaVision
- Other Tools : GitHub, Slack, Notion, Discord, GCS, GEE, MS word, PowerPoint

5 Awards

- **Provost's Award** in 2025 research challenge / StonyBrook University / November 2025
- **Second Place** in Graduation Project Competition / Incheon National University, Department of Computer Science / May 2024
- **First Place** in AI Competition / Incheon National University / November 2023